

Today's Agenda

1) Unified Modelling bqe-M3

2) Text-2-SQL (Llama)
✓ (Vaana.ai)

Supabase

COLBERT

(Contextualized late Interaction Over BERT)

$R(q, d)$

Query ✓

{ Neural Retrieval }

Embedding &
Ranking

{ Multi Vectors }

Max Sim Operation

Maximum Similarity

Multiple Contextualized Vectors

Query $\rightarrow$ Token level

Better Semantic

DENSE + Colbert

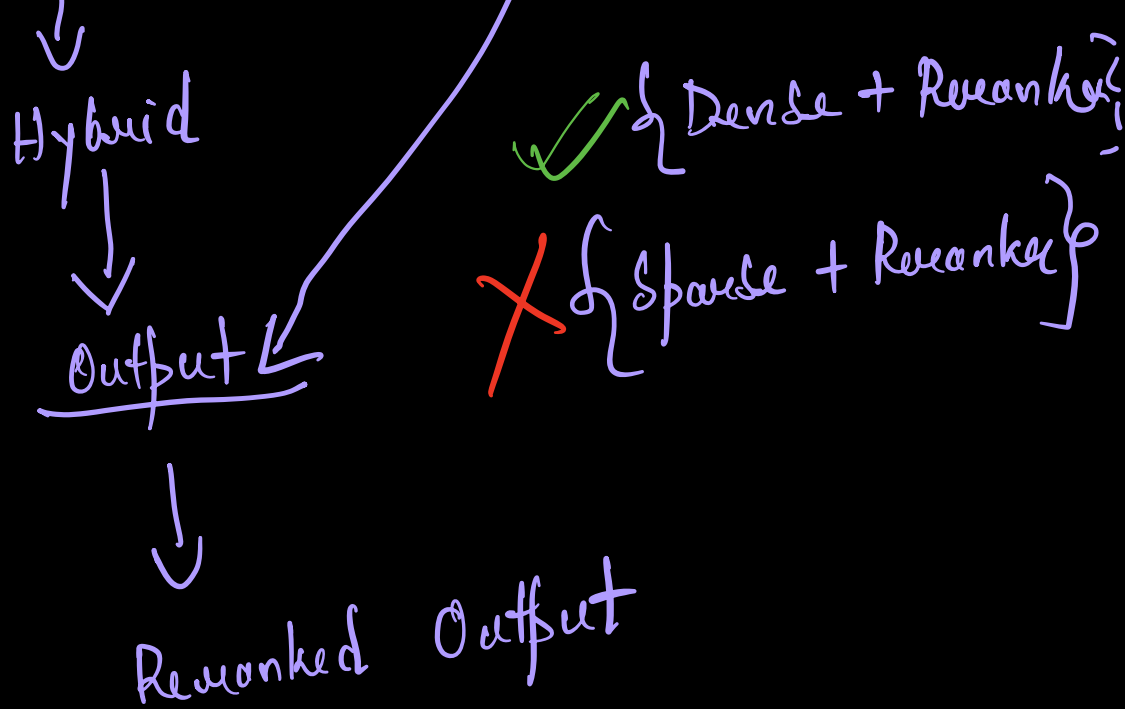
✓ RSF or RAF (Hybrid Search) ^{D/S}

Relevance (ms-marco-LM-v2)

COLBERT V2

2 ~ 3%

{ Dense, Sparse, Multivectors }



Second Stage Retrieval Systems

Re-rankers → Cross Encoder

Embedding Models → Bi-Encoder

BGE-M3 → Multi Vectors (Bi-Encoding)

BiEncoders are always faster than cross encoders.

COLBERT $\rightarrow$ Biencoders

Late Interaction

Bi Encoder

COLBERT

Cross Encode

fast, less accuracy

slow, high accuracy

Joint

Separate

Query - Doc Attention

N/A Biencoder

✓✓ Colbert

Vector Computing

Bi- Encoders

Query $\rightarrow$ E $\rightarrow$ [CLS] [Vector]

Doc $\rightarrow E \rightarrow [US] [vector]$

Score :- $E_q \cdot E_d$

COLBERT

Score :- $\text{MaxSim}(E_q, E_d)$

Cross Encoders :-

$R(q, d)$